Differentiation

Solutions

Chapter Overview

Solution 2

In Year 1 we differentiated terms of the form ax^n. This chapter extends those methods.

1. **Trigonometric, exponential and logarithmic functions**

Find $\frac{d}{dx}\sin x$, $\frac{d}{dx}\ln x$ and $\frac{d}{dx}2^x$.

2. **Chain, product and quotient rules**

Differentiate composite functions, products and quotients, such as $x\sin x$ and e^{2x^2}.

3. **Parametric differentiation**

If $x = \sin t$, $y = e^t$, find the tangent at $(0, 1)$.

4. **Implicit differentiation**

An implicit relationship need not give y explicitly in terms of x.

If $xy + y^2 = 3$, find $\frac{dy}{dx}$.

5. **Rates of change**

A circle's radius increases at 5 cm/s. How fast does its area change when $r = 3$ cm?

Differentiating Trigonometric Functions — Solution 3

Differentiate $\sin x$ from first principles.

Recall, as $h \to 0$, $\sin h \approx h$, $\cos h \approx 1 - \frac{1}{2}h^2$.

Also $\sin(x+h) = \sin x \cos h + \cos x \sin h$.

First principles and addition formula

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$
$$= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$$
$$= \lim_{h\to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}.$$

Collect the sine and cosine terms

$$\frac{dy}{dx} = \lim_{h\to 0} \left(\frac{\cos h - 1}{h} \sin x + \frac{\sin h}{h} \cos x \right).$$

Use the small-angle limits

$$\lim_{h\to 0} \frac{\sin h}{h} = 1, \qquad \lim_{h\to 0} \frac{\cos h - 1}{h} = 0.$$

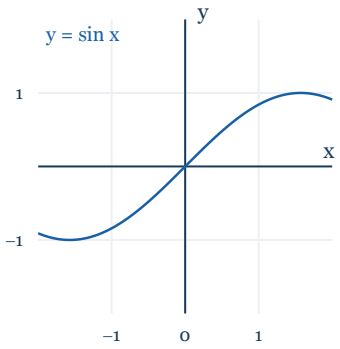
The approximations give $h/h = 1$ and $-h^2/(2h) = -h/2 \to 0$.

$$\frac{dy}{dx} = 0 \sin x + 1 \cos x = \cos x.$$

Why Must We Use Radians? — Solution 4

Explanation 1

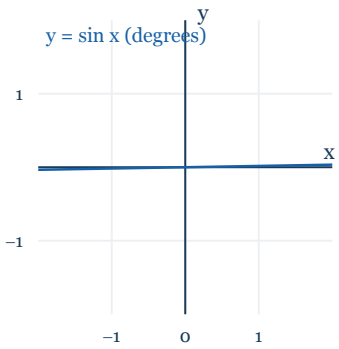
The approximations $\sin x \approx x$ and $\cos x \approx 1 - \frac{1}{2}x^2$, as $x \to 0$, require radians. The first-principles proof used these approximations.



In radians, $\cos 0 = 1$, so the predicted gradient at the origin is 1. This agrees with the graph.

Explanation 2

Compare the graphs with the same scale on both axes.



In degrees, the gradient at the origin is much smaller than 1. The formula $\frac{d}{dx}\sin x = \cos x$ therefore does not apply to degree inputs.

Differentiating Trigonometric Functions

Solution 5

Find $\frac{d}{dx}\cos x$.

$$\frac{d}{dx}\cos x = -\sin x.$$

Memorisation tip

Moving right differentiates; moving left integrates.

$\sin x \longrightarrow \cos x$

For the "wrong" function, the sign changes: differentiating $\cos x$, or integrating $\sin x$.

Differentiating Sine and Cosine of kx

Solution 6

Find $\frac{d}{dx}\sin kx$.

$k\cos kx$

Find $\frac{d}{dx}\cos kx$.

$-k\sin kx$

These results come from the chain rule, later in this chapter.

Quickfire questions

a. $\frac{d}{dx}\sin 3x$

a
$3\cos 3x$

b. $\frac{d}{dx}\cos 5x$

b
$-5\sin 5x$

c. $\frac{d}{dx}(3\sin 5x)$

c
$15\cos 5x$

d. $\frac{d}{dx}(4\cos 3x)$

d
$-12\sin 3x$

e. $\frac{d}{dx}(-\frac{1}{2}\sin x)$

e
$-\frac{1}{2}\cos x$

f. $\frac{d}{dx}(-\frac{2}{3}\cos \frac{x}{2})$

f
$\frac{1}{3}\sin \frac{x}{2}$

Example — Stationary Points

Solution 7

A curve has equation $y = \frac{1}{2}x - \cos 2x$. Find its stationary points for $0 \leq x \leq \pi$.

Working

$$\frac{dy}{dx} = \frac{1}{2} + 2\sin 2x = 0$$

$$2\sin 2x = -\frac{1}{2}, \qquad \sin 2x = -\frac{1}{4}.$$

The principal angle is $-0.2527\ldots$. In $0 \leq 2x \leq 2\pi$:

$2x = 3.3943\ldots,\ 6.0305\ldots$

$x = 1.697\ldots,\ 3.015\ldots$.

Substitute into $y = \frac{1}{2}x - \cos 2x$:

$y = 1.82\ldots,\ 0.539\ldots$.

Stationary points: $(1.697, 1.82)$ and $(3.015, 0.539)$, approximately.

$y = \frac{1}{2}(1.697) - \cos(2 \times 1.697) \approx 1.82$.

Test Your Understanding — Stationary Points

Solution 8

A curve has equation $y = \sin 3x + 2x$. Find its stationary points for $0 \leq x \leq \frac{2}{3}\pi$.

Working

$$\frac{dy}{dx} = 3\cos 3x + 2 = 0$$

$$3\cos 3x = -2, \qquad \cos 3x = -\frac{2}{3}.$$

$3x = 2.3005\ldots,\ 3.9827\ldots$

$x = 0.7668\ldots,\ 1.3276\ldots$.

Substituting into $y = \sin 3x + 2x$ gives $y = 2.279\ldots$ and $1.910\ldots$.

Stationary points: $(0.7668, 2.279)$ and $(1.3276, 1.910)$, approximately.

Exercise 9A — Page 234 Solution 9

1. a. Given $f(x) = \cos x$, show that

$$f\prime(x) = \lim_{h\to 0}\left(\frac{\cos h - 1}{h}\cos x - \frac{\sin h}{h}\sin x\right).$$

b. Hence prove that $f\prime(x) = -\sin x$.

1a

$$f\prime(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h}$$
$$= \lim_{h\to 0}\frac{\cos(x+h) - \cos x}{h}$$
$$= \lim_{h\to 0}\frac{\cos x\cos h - \sin x\sin h - \cos x}{h}$$
$$= \lim_{h\to 0}\left(\frac{\cos h - 1}{h}\cos x - \frac{\sin h}{h}\sin x\right).$$

1b

As $h \to 0$,

$$\frac{\cos h - 1}{h} \to 0, \qquad \frac{\sin h}{h} \to 1.$$

Thus

$$f\prime(x) = 0\cos x - 1\sin x = -\sin x.$$

Exercise 9A — Page 234 Solution 10

7. A curve has equation $y = 2\sin 2x + \cos 2x$. Find its stationary points for $0 \le x \le \pi$.

7

$$\frac{dy}{dx} = 2(2\cos 2x) - 2\sin 2x = 4\cos 2x - 2\sin 2x.$$

At stationary points, $dy/dx = 0$:

$$4\cos 2x - 2\sin 2x = 0$$

$$4 - 2\tan 2x = 0, \qquad \tan 2x = 2.$$

$$2x = 1.107\ldots,\ 4.248\ldots$$

$$x = 0.554\ldots,\ 2.12\ldots.$$

Substituting into $y = 2\sin 2x + \cos 2x$ gives

$$y = 2.24\ldots,\ -2.24\ldots.$$

The points are $(0.554, 2.24)$ and $(2.12, -2.24)$ to 3 s.f.

$$y = 2\sin(2\times 0.554) + \cos(2\times 0.554) \approx 2.24,$$
$$y = 2\sin(2\times 2.12) + \cos(2\times 2.12) \approx -2.24.$$

Exercise 9A — Page 234

Solution 11

8. A curve has equation $y = \sin 5x + \cos 3x$. Find the equation of its tangent at $(\pi, -1)$. **(4 marks)**

8

$\dfrac{dy}{dx} = 5\cos 5x - 3\sin 3x.$

At $(\pi, -1)$:

$\dfrac{dy}{dx} = 5\cos 5\pi - 3\sin 3\pi$

$= 5(-1) - 3(0) = -5.$

The tangent is

$y - (-1) = -5(x - \pi),$

$y = -5x + 5\pi - 1.$

Exercise 9A — Page 234

Solution 12

9. A curve has equation $y = 2x^2 - \sin x$. Show that its normal at $x = \pi$ has equation $x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0$. **(7 marks)**

9

$\dfrac{dy}{dx} = 4x - \cos x.$

At $x = \pi$, $y = 2\pi^2$ and $dy/dx = 4\pi + 1$.

The normal gradient is $-1/(4\pi + 1)$, so

$y - 2\pi^2 = -\dfrac{1}{4\pi + 1}(x - \pi).$

Multiply by $4\pi + 1$ and rearrange:

$x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0.$

Exercise 9A — Page 234

Solution 13

10. Prove from first principles that the derivative of $\sin x$ is $\cos x$.

You may use the addition formula for $\sin(A+B)$, and that as $h \to 0$,

$$\frac{\sin h}{h} \to 1, \qquad \frac{\cos h - 1}{h} \to 0.$$

10

Let $f(x) = \sin x$. Then

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h\to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$$

$$= \lim_{h\to 0} \left(\frac{\cos h - 1}{h} \sin x + \frac{\sin h}{h} \cos x \right).$$

Using the given limits,

$$f'(x) = 0 \sin x + 1 \cos x = \cos x.$$

Differentiating Exponentials and Logarithms

Solution 14

Recall from Year 1

$$\frac{d}{dx} e^x = e^x.$$

Just seen

$$\frac{d}{dx} \sin kx = k \cos kx.$$

Applying the same input-scaling idea $\longrightarrow$

$$\frac{d}{dx} e^{kx} = k e^{kx}.$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} a^x = a^x \ln a$$

$$\frac{d}{dx} a^{kx} = k a^{kx} \ln a$$

For positive k, x,

$$\frac{d}{dx} \ln(kx) = \frac{d}{dx}(\ln k + \ln x) = \frac{1}{x}.$$

Scaling the logarithm's input does not change the gradient. Proofs follow using implicit differentiation.

Quickfire Questions — 1/2

Solution 15

1. $\frac{d}{dx}(3^x)$

1 $3^x \ln 3$

2. $\frac{d}{dx}(x^3)$

2 $3x^2$

3. $\frac{d}{dx}(\ln(3x))$

3 $\frac{1}{x}$

4. $\frac{d}{dx}(3^{2x})$

4 $3^{2x}(2 \ln 3)$

5. $\frac{d}{dx}(2^{3x})$

5 $2^{3x}(3 \ln 2)$

6. $\frac{d}{dx}(5 \ln x)$

6 $\frac{5}{x}$

7. $\frac{d}{dx}(e^{x/2})$

7 $\frac{1}{2}e^{x/2}$

8. $\frac{d}{dx}(5 \ln(2x))$

8 $\frac{5}{x}$

9. $\frac{d}{dt}(9^t)$

9 $9^t \ln 9$

10. $\frac{d}{dx}(5(4^x))$

10 $5(4^x) \ln 4$

11. $\frac{d}{dx}(x^4)$

11 $4x^3$

12. $\frac{d}{dx}(\ln(6x))$

12 $\frac{1}{x}$

13. $\frac{d}{dx}(6 \ln x)$

13 $\frac{6}{x}$

14. $\frac{d}{dx}(3e^{2x})$

14 $6e^{2x}$

15. $\frac{d}{dx}(e^{-x})$

15 $-e^{-x}$

Quickfire Questions — 2/2

Solution 16

Applied example

$P = 1000(2^t)$ models a rabbit population after t years. When does its rate of increase reach 20,000 rabbits per year?

Application

$$\frac{dP}{dt} = 1000(2^t) \ln 2 = 20000$$

$$2^t = \frac{20000}{1000 \ln 2} = \frac{20}{\ln 2}$$

$$t = \log_2\left(\frac{20}{\ln 2}\right) = 4.85 \text{ years (3 s.f.).}$$

Test Your Understanding — Solution 17

Differentiate $y = (e^x + 2)^2$.

Hint: Expand first.

Derivative

$y = e^{2x} + 4e^x + 4$

$\dfrac{dy}{dx} = 2e^{2x} + 4e^x.$

A child's headlice are treated with shampoo. The population after t days is modelled by $P = 460(3^{-2t})$.

a. When are 20 headlice left?

b. Find the rate of change after 3 days.

a

$20 = 460(3^{-2t})$

$3^{-2t} = \dfrac{1}{23}$

$-2t = \log_3(1/23) = -2.85\ldots$

$t = 1.43$ days (3 s.f.).

b

$\dfrac{dP}{dt} = -920(3^{-2t}) \ln 3.$

At $t = 3$:

$\dfrac{dP}{dt} = -920(3^{-6}) \ln 3$

$= -1.39$ headlice/day (3 s.f.).

Exercise 9B — Pages 236–237 — Solution 18

4. Find the tangent to $y = 2^x + 2^{-x}$ at $(2, \frac{17}{4})$.

4

$\dfrac{dy}{dx} = 2^x \ln 2 - 2^{-x} \ln 2.$

At $x = 2$:

$\dfrac{dy}{dx} = 4 \ln 2 - \dfrac{1}{4} \ln 2 = \dfrac{15}{4} \ln 2.$

Thus

$y - \dfrac{17}{4} = \dfrac{15}{4} \ln 2\,(x - 2),$

$4y = (15 \ln 2)x + (17 - 30 \ln 2).$

Exercise 9B — Pages 236–237 Solution 19

5. A curve has equation $y = e^{2x} - \ln x$. Show that the tangent at $x = 1$ is
$y = (2e^2 - 1)x - e^2 + 1$.
(6 marks)

5

$$\frac{dy}{dx} = 2e^{2x} - \frac{1}{x}.$$

At $x = 1$, $y = e^2$ and $dy/dx = 2e^2 - 1$.

The tangent is

$$y - e^2 = (2e^2 - 1)(x - 1).$$

Rearranging,

$$y = (2e^2 - 1)x - 2e^2 + 1 + e^2$$
$$= (2e^2 - 1)x - e^2 + 1.$$

Exercise 9B — Pages 236–237 Solution 20

7. Cambridge's population was 37,000 in 1900 and about 109,000 in 2000. Model the population t years after 1900 by $P = P_0 k^t$.

a. Find P_0 and k.

b. Evaluate dP/dt in 2000.

c. Interpret your answer.

7a

At $t = 0$,

$$37000 = P_0 k^0 = P_0.$$

At $t = 100$,

$$109000 = 37000k^{100}$$
$$\frac{109}{37} = k^{100}$$
$$k = \left(\frac{109}{37}\right)^{1/100} = 1.01086287\ldots$$
$$k = 1.01 \text{ (2 d.p.)}.$$

7b

$$P = P_0 k^t \implies \frac{dP}{dt} = P_0 k^t \ln k.$$

Using the unrounded value of k, at $t = 100$:

$$\frac{dP}{dt} = 37000k^{100} \ln k \approx 1178.$$
$$37000\left[(109/37)^{1/100}\right]^{100} \ln\left[(109/37)^{1/100}\right] \approx 1178.$$

7c

The model predicts a population increase of approximately 1,178 people per year in 2000.

Exercise 9B — Pages 236–237 Solution 21

9. Prove that $\frac{d}{dx}a^{kx} = ka^{kx}\ln a$, assuming $\frac{d}{dx}e^{kx} = ke^{kx}$.

9

Let $y = a^{kx}$. Then

$y = e^{\ln(a^{kx})} = e^{kx\ln a} = e^{(k\ln a)x}$.

$\dfrac{dy}{dx} = (k\ln a)e^{(k\ln a)x}$

$= k\ln a\, e^{kx\ln a} = ka^{kx}\ln a.$

10. $f(x) = e^{2x} - \ln(x^2) + 4$, $x > 0$.

a. Find $f\prime(x)$.

b. The curve has gradient 2 at $x = a$. Show that $a(e^{2a} - 1) = 1$.

10a

$f(x) = e^{2x} - 2\ln x + 4$

$f\prime(x) = 2e^{2x} - \dfrac{2}{x}.$

10b

$2e^{2a} - \dfrac{2}{a} = 2$

$e^{2a} - \dfrac{1}{a} - 1 = 0$

$ae^{2a} - 1 - a = 0$

$a(e^{2a} - 1) = 1.$

Exercise 9B — Pages 236–237 Solution 22

11. Curve C has equation $y = 5\sin 3x + 2\cos 3x$, $-\pi \le x \le \pi$.

a. Show that $P(0, 2)$ lies on C.

b. Find the normal at P.

11a

At $x = 0$:

$y = 5\sin 0 + 2\cos 0 = 0 + 2 = 2.$

Hence $P(0, 2)$ lies on the curve.

11b

$\dfrac{dy}{dx} = 15\cos 3x - 6\sin 3x.$

At $x = 0$, $dy/dx = 15\cos 0 - 6\sin 0 = 15.$

Therefore the normal is

$y - 2 = -\dfrac{1}{15}(x - 0), \qquad y = -\dfrac{x}{15} + 2.$

Exercise 9B — Pages 236–237

Solution 23

12. Point P lies on $y = 2(3^{4x})$ with $x = 1$. Find the normal at P in the form $y = ax + b$, giving a, b exactly.

12

$$\frac{dy}{dx} = 2(3^{4x})(4\ln 3) = 8\ln 3\,(3^{4x}).$$

At $x = 1$,

$$y = 2(81) = 162, \qquad \frac{dy}{dx} = 8\ln 3\,(3^4) = 648\ln 3.$$

The normal is

$$y - 162 = -\frac{1}{648\ln 3}(x - 1),$$

$$y = -\frac{x}{648\ln 3} + \frac{1}{648\ln 3} + 162.$$

For Your Interest — Differentiating ln x

Solution 24

Prove from first principles that $\frac{d}{dx}\ln x = \frac{1}{x}$, for $x > 0$.

First principles; then the logarithm law

$$\frac{dy}{dx} = \lim_{h\to 0}\frac{\ln(x+h) - \ln x}{h}$$

$$= \lim_{h\to 0}\frac{\ln((x+h)/x)}{h}$$

$$= \lim_{h\to 0}\frac{1}{h}\ln\left(1 + \frac{h}{x}\right).$$

Move the multiplier into the power

$$= \lim_{h\to 0}\ln\left(1 + \frac{h}{x}\right)^{1/h}.$$

Substitute $n = 1/h$

For the right-hand limit, $n \to \infty$:

$$\lim_{n\to\infty}\ln\left(1 + \frac{1/n}{x}\right)^{n}$$

$$= \ln\left(\lim_{n\to\infty}\left(1 + \frac{1/x}{n}\right)^{n}\right).$$

Here $(1/n)/x = (1/x)/n$. Continuity of ln lets us move the limit inside.

Using $e^k = \lim_{n\to\infty}(1 + k/n)^n$, the result is $\ln(e^{1/x}) = 1/x$. The left-hand limit agrees by the same exponential limit.

Hence $\frac{dy}{dx} = \frac{1}{x}$.

Differentiating Combinations of Functions

Solution 25

Functions can interact in different ways. Choose the matching rule.

1. **Composite function** $\longrightarrow$ **Chain rule**

$y = f(g(x)) = \sqrt{1+3x}$.

Outer function: $f(x) = \sqrt{x}$.

Inner function: $g(x) = 1 + 3x$.

Exercise 9C

2. **Product of two functions** $\longrightarrow$ **Product rule**

$y = f(x)g(x) = x \sin 2x$.

Exercise 9D

3. **Quotient of two functions** $\longrightarrow$ **Quotient rule**

$y = f(x)/g(x) = \ln x/x$.

Exercise 9E

The Chain Rule

Solution 26

The chain rule differentiates a composite function: a function within a function.

Example: $y = (3x^4 + x)^5$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$

Note

The du symbols appear to cancel. This is a memory aid, not a proof.

Full method

Let $u = 3x^4 + x$: this is the inner function.

$$\frac{du}{dx} = 12x^3 + 1.$$

The outer function is $y = u^5$, so

$$\frac{dy}{du} = 5u^4 = 5(3x^4 + x)^4.$$

$$\frac{dy}{dx} = 5(3x^4 + x)^4(12x^3 + 1).$$

Mentally in one go

Differentiate the outer function, keeping its input unchanged, then multiply by the derivative of the inner function.

Think “something to the fifth power”: its outer derivative is five times that same “something” to the fourth power.

Outer derivative: $5(3x^4 + x)^4$.

Now differentiate the inner “something”, $3x^4 + x$.

Inner derivative: $12x^3 + 1$.

Multiply:

$$\frac{dy}{dx} = 5(3x^4 + x)^4(12x^3 + 1).$$

Practice — Chain Rule — 1/2

Solution 27

Original function	Outer derivative	Inner derivative
1. $y = (x^2 + 1)^3$	1 outer $3(x^2 + 1)^2$ Differentiate the cube.	1 inner $2x$ Multiply by the derivative of the inside function.
2. $y = (\ln x)^3$	2 outer $3(\ln x)^2$ Differentiate the cube.	2 inner $1/x$ Multiply by the derivative of the inside function.
3. $y = e^{x^2+x}$	3 outer e^{x^2+x} Differentiate the exponential.	3 inner $2x + 1$ Multiply by the derivative of the exponent.

Practice — Chain Rule — 2/2

Solution 28

Original function	Outer derivative	Inner derivative
4. $y = (2^x + 1)^2$	4 outer $2(2^x + 1)$ Differentiate the square.	4 inner $2^x \ln 2$ Multiply by the derivative of the inside function.
5. $y = \ln(\sin x)$	5 outer $1/\sin x$ Differentiate the natural logarithm.	5 inner $\cos x$ Multiply by the derivative of the inside function.

Further Practice — Rewrite Before Differentiating — 1/2

Solution 29

1. $y = \sin 5x$

1 outer
$\cos 5x$

Differentiate the sine.

1 inner
5

Differentiate the inside function; multiply the factors.

2. $y = \sin^2 x$

2 rewrite $y = (\sin x)^2$

Rewrite using brackets first.

2 outer $2 \sin x$

Differentiate the square.

2 inner $\cos x$

Differentiate the inside function; multiply the factors.

3. $y = \sqrt{x+1}$

3 rewrite $y = (x+1)^{1/2}$

Rewrite using brackets first.

3 outer $\frac{1}{2}(x+1)^{-1/2}$

Differentiate the power of one half.

3 inner 1

Differentiate the inside function; multiply the factors.

Further Practice — Rewrite Before Differentiating — 2/2

Solution 30

4. $y = \cos^3 2x$

4 rewrite
$y = (\cos 2x)^3$

Rewrite using brackets first.

4 outer $3(\cos 2x)^2$

Differentiate the cube.

4 inner $-2 \sin 2x$

Use the chain rule twice.

5. $y = e^{e^x}$

5 outer
e^{e^x}

Differentiate the exponential.

5 inner
e^x

Differentiate the inside function; multiply the factors.

Test Your Understanding — Chain Rule

Solution 31

C3 June 2011, Q1a (2 marks)

Differentiate $\ln(x^2 + 3x + 5)$.

Exam

$$\frac{dy}{dx} = \frac{1}{x^2 + 3x + 5}(2x + 3)$$
$$= \frac{2x + 3}{x^2 + 3x + 5}.$$

In general

$$\frac{d}{dx}\ln(f(x)) = \frac{f\prime(x)}{f(x)}.$$

Textbook example

Given $y = \sqrt{5x^2 + 1}$, find dy/dx at $(4, 9)$.

Textbook

$$y = (5x^2 + 1)^{1/2}$$
$$\frac{dy}{dx} = \frac{1}{2}(5x^2 + 1)^{-1/2}(10x).$$

At $x = 4$:

$$\frac{dy}{dx} = \frac{1}{2}(5 \cdot 4^2 + 1)^{-1/2}(10 \cdot 4)$$
$$= \frac{20}{9}.$$

Finding dy/dx from dx/dy

Solution 32

Sometimes x is given in terms of y, but we want dy/dx.

$$\frac{dy}{dx} = \frac{1}{dx/dy}$$

Proof using the chain rule

$$\frac{dy}{dy} = \frac{dy}{dx}\frac{dx}{dy}.$$

Since $dy/dy = 1$, divide by $dx/dy \neq 0$.

Example 1

Find dy/dx when $x = 2y^2 + y$.

$$\frac{dx}{dy} = 4y + 1, \qquad \frac{dy}{dx} = \frac{1}{4y + 1}.$$

Example 2

Find the gradient of $x = (1 + 2y)^3$ when $y = 1$.

$$\frac{dx}{dy} = 6(1 + 2y)^2$$
$$\frac{dy}{dx} = \frac{1}{6(1 + 2y)^2}.$$

At $y = 1$:

$$m = \frac{1}{6(1 + 2)^2} = \frac{1}{54}.$$

Exercise 9C — Pages 239–240

Solution 33

4. Curve C has equation $y = (5 - 2x)^3$. Find its tangent at the point P where $x = 1$.

4

Let $u = 5 - 2x$, so $y = u^3$.

$$\frac{du}{dx} = -2, \qquad \frac{dy}{du} = 3u^2.$$

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = 3u^2(-2) = -6(5 - 2x)^2.$$

When $x = 1$, $y = 3^3 = 27$ and $dy/dx = -6 \cdot 3^2 = -54$.

The tangent at $P(1, 27)$ is

$$y - 27 = -54(x - 1), \qquad y = -54x + 81.$$

Exercise 9C — Pages 239–240

Solution 34

5. Given $y = (1 + \ln 4x)^{3/2}$, find dy/dx at $x = \frac{1}{4}e^3$.

5

Let $u = 1 + \ln 4x$, $y = u^{3/2}$.

$$\frac{du}{dx} = \frac{1}{x}, \quad \frac{dy}{du} = \frac{3}{2}u^{1/2}.$$

$$\frac{dy}{dx} = \frac{3}{2}u^{1/2}\frac{1}{x} = \frac{3}{2x}\sqrt{1 + \ln 4x}.$$

At $x = \frac{1}{4}e^3$:

$$\frac{dy}{dx} = \frac{3}{\frac{1}{2}e^3}\sqrt{1 + \ln e^3}$$

$$= \frac{6}{e^3}\sqrt{4} = 12e^{-3}.$$

9. a. Differentiate $e^y = x$ with respect to y.

b. Hence prove that $y = \ln x$ gives $dy/dx = 1/x$.

9a

$$x = e^y \implies \frac{dx}{dy} = e^y.$$

9b

$$y = \ln x \implies e^y = x.$$

$$\frac{dy}{dx} = \frac{1}{dx/dy} = \frac{1}{e^y} = \frac{1}{x}.$$

Exercise 9C — Pages 239–240 Solution 35

10. Curve C has equation $x = 4\cos 2y$.

a. Show that $Q(2, \pi/6)$ lies on C.

b. Show that $dy/dx = -1/(4\sqrt{3})$ at Q.

c. Find the normal at Q, in the form $ax + by + c = 0$, with exact constants.

10a

At $x = 2$, $\cos 2y = 1/2$.

$2y = \pi/3,\ 5\pi/3, \ldots$

$y = \pi/6,\ 5\pi/6, \ldots$.

Hence $Q(2, \pi/6)$ lies on C.

10b

$$\frac{dx}{dy} = -8\sin 2y, \quad \frac{dy}{dx} = -\frac{1}{8\sin 2y}.$$

At Q:

$$\frac{dy}{dx} = -\frac{1}{8\sin(\pi/3)} = -\frac{1}{4\sqrt{3}}.$$

10c

The normal has gradient $4\sqrt{3}$:

$$y - \frac{\pi}{6} = 4\sqrt{3}(x - 2),$$

$$4\sqrt{3}x - y - 8\sqrt{3} + \frac{\pi}{6} = 0.$$

Exercise 9C — Pages 239–240 Solution 36

12. Curve C has equation $y = 4/(2 - 4x)^2$, $x \neq 1/2$. Point A has $x = 3$. Find the normal at A in the form $ax + by + c = 0$, with integer constants.

12

Let $u = 2 - 4x$, so $y = 4u^{-2}$.

$$\frac{du}{dx} = -4, \quad \frac{dy}{du} = -8u^{-3}.$$

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \frac{32}{(2 - 4x)^3}.$$

At $x = 3$, $y = 4/(-10)^2 = 0.04$ and $dy/dx = 32/(-10)^3 = -0.032$.

The normal is

$$y - 0.04 = \frac{1}{0.032}(x - 3).$$

Multiply by 100 and rearrange:

$$100y - 4 = 3125x - 9375,$$

$$3125x - 100y - 9371 = 0.$$

Extension — STEP I 2014, Q4

Solution 37

[STEP I 2014 Q4] An accurate clock has an hour hand of length a and a minute hand of length b (where $b > a$), both measured from the pivot at the centre of the clock face. Let x be the distance between the ends of the hands when the angle between the hands is θ, where $0 \leq \theta < \pi$.

Show that the rate of increase of x is greatest when $x = \left(b^2 - a^2\right)^{\frac{1}{2}}$.

In the case when $b = 2a$ and the clock starts at midday (with both hands pointing vertically upwards), show that this occurs for the first time a little less than 11 minutes later.

Setup

By the cosine rule, $x^2 = a^2 + b^2 - 2ab\cos\theta$.

$$\frac{dx}{d\theta} = \frac{ab\sin\theta}{\sqrt{a^2 + b^2 - 2ab\cos\theta}}.$$

The angular speed is constant, so maximising dx/dt is equivalent to maximising $dx/d\theta$.

Maximum

Let $c = \cos\theta$. Differentiating again and setting the numerator to zero gives

$abc^2 - (a^2 + b^2)c + ab = 0$

$(bc - a)(ac - b) = 0.$

Only $c = a/b$ lies in $[-1, 1]$, since $b > a > 0$. The rate is positive inside $(0, \pi)$ and zero at both ends, so this is its maximum.

$x^2 = a^2 + b^2 - 2ab(a/b) = b^2 - a^2,$

$x = \sqrt{b^2 - a^2}.$

Time

When $b = 2a$, $\cos\theta = 1/2$, so $\theta = \pi/3$.

Relative angular speed is $2\pi - \pi/6 = 11\pi/6$ radians/hour.

$$t = \frac{\pi/3}{11\pi/6} = \frac{2}{11} \text{ hour}$$

$$= \frac{120}{11} \text{ minutes} \approx 10.91 \text{ minutes}.$$

The Product Rule

Solution 38

Use the product rule for a product of two functions:

$$y = uv \implies \frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}.$$

Method tips

Write out $u, v, u\prime, v\prime$: the individual derivatives can be complicated. Factorising is typical when finding turning points. Exponentials cannot equal zero.

Memorisation tip

1. Differentiate one factor; leave the other.
2. Do it the other way round.
3. Add. Either order gives the same result.

Example 1

Find dy/dx for $y = x^2 \sin x$.

Write u, v, then differentiate each.

1 factors $u = x^2, \quad v = \sin x, \quad u\prime = 2x, \quad v\prime = \cos x.$

Substitute into the product rule.

1 result

$$\frac{dy}{dx} = x^2\cos x + 2x\sin x.$$

Example 2

Find the turning point of $y = xe^{2x}$.

2 derivative

$u = x, \ v = e^{2x}, \quad u\prime = 1, \ v\prime = 2e^{2x}.$

$$\frac{dy}{dx} = 2xe^{2x} + e^{2x}.$$

2 x

Set $dy/dx = 0$: $e^{2x}(2x + 1) = 0$.

Since $e^{2x} \neq 0$, $2x + 1 = 0$, so $x = -1/2$.

2 y

Substitute: $y = -\frac{1}{2}e^{-1} = -\frac{1}{2e}$.

Turning point: $(-\frac{1}{2}, -\frac{1}{2e})$.

Product and Chain Rule Examples

Solution 39

Example 3

If $y = e^{4x}\sin^2 3x$, show that

$\dfrac{dy}{dx} = e^{4x}\sin 3x(A\cos 3x + B\sin 3x)$,

finding A, B.

$u = e^{4x}, \quad v = (\sin 3x)^2, \quad u\prime = 4e^{4x}.$

$v\prime = 2\sin 3x(3\cos 3x) = 6\sin 3x\cos 3x.$

$\dfrac{dy}{dx} = 6e^{4x}\sin 3x\cos 3x + 4e^{4x}(\sin 3x)^2$

$= e^{4x}\sin 3x(6\cos 3x + 4\sin 3x).$

$A = 6,\ B = 4.$

Example 4

Find $f\prime(x)$ for $f(x) = x^2\sqrt{3x-1}$.

$u = x^2, \quad v = (3x-1)^{1/2}, \quad u\prime = 2x,$

$v\prime = \tfrac{1}{2}(3x-1)^{-1/2}(3) = \tfrac{3}{2}(3x-1)^{-1/2}.$

$f\prime(x) = \tfrac{3}{2}x^2(3x-1)^{-1/2} + 2x(3x-1)^{1/2}$

$= \tfrac{1}{2}x(3x-1)^{-1/2}[3x + 4(3x-1)]$

$= \tfrac{1}{2}x(3x-1)^{-1/2}(15x-4) = \dfrac{x(15x-4)}{2\sqrt{3x-1}}.$

Factoring tip

Take out the smallest power, $-1/2$. The other power is one higher, leaving $(3x-1)$. Also take out $1/2$, common to $3/2$ and 2.

The final radical form is optional, but avoids negative powers.

Test Your Understanding — Product Rule

Solution 40

Exam-style question 1 (4 marks)

Differentiate $x^2\ln(3x)$, simplifying your answer.

1

$u = x^2, \quad v = \ln(3x), \quad u\prime = 2x, \quad v\prime = \dfrac{1}{x}.$

$\dfrac{dy}{dx} = x^2\dfrac{1}{x} + 2x\ln(3x)$

$= x + 2x\ln(3x) = x[1 + 2\ln(3x)].$

Question 2

Given $\frac{dy}{dx} = \frac{1}{6x(x-1)^{1/2}}$, find d^2y/dx^2, simplifying your answer.

2

$\dfrac{dy}{dx} = \tfrac{1}{6}x^{-1}(x-1)^{-1/2}.$

$\dfrac{d^2y}{dx^2} = -\tfrac{1}{6}x^{-2}(x-1)^{-1/2} - \tfrac{1}{12}x^{-1}(x-1)^{-3/2}$

$= -\tfrac{1}{12}x^{-2}(x-1)^{-3/2}[2(x-1) + x]$

$= \dfrac{2-3x}{12x^2(x-1)^{3/2}}.$

Examiner comment

“Perhaps the most challenging question in the paper.”

Exercise 9D — Page 242

Solution 41

6. Curve C: $y = x^2\cos(x^2)$. Find the tangent at $P(\sqrt{\pi}/2, \pi\sqrt{2}/8)$ in the form $ax + by + c = 0$, with exact constants. **(7 marks)**

6 derivative

$u = x^2, \quad v = \cos(x^2).$

$u\prime = 2x, \quad v\prime = -2x\sin(x^2).$

$$\frac{dy}{dx} = x^2[-2x\sin(x^2)] + 2x\cos(x^2)$$

$$= 2x[\cos(x^2) - x^2\sin(x^2)].$$

At $x = \sqrt{\pi}/2$:

$$m = \sqrt{\pi}\left(\cos\frac{\pi}{4} - \frac{\pi}{4}\sin\frac{\pi}{4}\right)$$

$$= \sqrt{\pi}\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4\sqrt{2}}\right) = \sqrt{\frac{\pi}{2}}\left(1 - \frac{\pi}{4}\right).$$

6 tangent

The tangent is

$$y - \frac{\pi\sqrt{2}}{8} = \sqrt{\frac{\pi}{2}}\left(1 - \frac{\pi}{4}\right)\left(x - \frac{\sqrt{\pi}}{2}\right).$$

$$8y - \pi\sqrt{2} = 4\sqrt{2\pi}(1 - \pi/4)(x - \sqrt{\pi}/2).$$

$$8y - \pi\sqrt{2} = \sqrt{2\pi}(4 - \pi)\left(x - \frac{\sqrt{\pi}}{2}\right)$$

$$= \sqrt{2\pi}(4 - \pi)x - \frac{\pi\sqrt{2}}{2}(4 - \pi).$$

$$\sqrt{2\pi}(\pi - 4)x + 8y + \tfrac{\pi\sqrt{2}}{2}(4 - \pi) - \pi\sqrt{2} = 0.$$

Hence

$$\sqrt{2\pi}(\pi - 4)x + 8y - \pi\sqrt{2}\left(\frac{\pi}{2} - 1\right) = 0.$$

$$a = \sqrt{2\pi}(\pi - 4),\ b = 8,\ c = -\pi\sqrt{2}(\pi/2 - 1).$$

Exercise 9D — Page 242

Solution 42

7. Given $y = 3x^2(5x - 3)^3$, show that $\dfrac{dy}{dx} = Ax(5x - 3)^n(Bx + C)$, finding A, n, B, C.

7

Let $u = 3x^2$, $v = (5x - 3)^3$.

$u\prime = 6x, \qquad v\prime = 15(5x - 3)^2.$

$$\frac{dy}{dx} = uv\prime + vu\prime$$

$$= 3x^2[15(5x - 3)^2] + 6x(5x - 3)^3$$

$$= 3x(5x - 3)^2[15x + 2(5x - 3)]$$

$$= 3x(5x - 3)^2(25x - 6).$$

$$A = 3,\ n = 2,\ B = 25,\ C = -6.$$

Exercise 9D — Page 242 — Solution 43

8. Curve C: $y = (x+3)^2 e^{3x}$.

a. Find dy/dx using the product rule.

b. Find the gradient at $x = 2$.

8a

$u = (x+3)^2, \quad v = e^{3x},$

$u\prime = 2(x+3), \quad v\prime = 3e^{3x}.$

$$\frac{dy}{dx} = (x+3)^2(3e^{3x}) + e^{3x}[2(x+3)]$$

$$= e^{3x}(x+3)[3(x+3)+2]$$

$$= e^{3x}(x+3)(3x+11).$$

8b

At $x = 2$:

$$\frac{dy}{dx} = e^6(5)(17) = 85e^6.$$

Exercise 9D — Page 242 — Solution 44

9. a. Differentiate $(2\sin x - 3\cos x)\ln 3x$.

9a

Let $u = 2\sin x - 3\cos x$, $v = \ln 3x$.

$u\prime = 2\cos x + 3\sin x, \quad v\prime = \dfrac{1}{x}.$

Using $dy/dx = uv\prime + vu\prime$:

$$\frac{dy}{dx} = \frac{2\sin x - 3\cos x}{x} + (2\cos x + 3\sin x)\ln 3x.$$

9. b. Differentiate $x^4 e^{7x-3}$.

9b

Let $u = x^4$, $v = e^{7x-3}$.

$u\prime = 4x^3, \quad v\prime = 7e^{7x-3}.$

$$\frac{dy}{dx} = x^4(7e^{7x-3}) + 4x^3 e^{7x-3}$$

$$= x^3 e^{7x-3}(7x+4).$$

Exercise 9D — Page 242

Solution 45

10. Find dy/dx at $x = 1$ on $y = x^5\sqrt{10x+6}$.

10

Let $u = x^5$, $v = (10x+6)^{1/2}$.

$u\prime = 5x^4, \quad v\prime = 10 \cdot \frac{1}{2}(10x+6)^{-1/2} = \dfrac{5}{\sqrt{10x+6}}.$

$\dfrac{dy}{dx} = uv\prime + vu\prime = \dfrac{5x^5}{\sqrt{10x+6}} + 5x^4\sqrt{10x+6}.$

At $x = 1$:

$\dfrac{dy}{dx} = \dfrac{5}{\sqrt{16}} + 5\sqrt{16} = 21.25.$

The Quotient Rule

Solution 46

Use the quotient rule to differentiate a quotient, just as the product rule differentiates a product.

Begin with $y = u/v$.

$$y = uv^{-1}$$

$$\frac{dy}{dx} = -uv^{-2}\frac{dv}{dx} + \frac{du}{dx}v^{-1}$$

$$= v^{-2}\left(v\frac{du}{dx} - u\frac{dv}{dx}\right)$$

$$= \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

The quotient rule

$$y = \frac{u}{v} \implies \frac{dy}{dx} = \frac{vu\prime - uv\prime}{v^2}.$$

Memorisation tip: "Bottoms first!"

The old denominator v comes first in the numerator; square it in the new denominator.

Note the subtraction in the numerator, unlike the addition in the product rule.

Examples — Quotient Rule

Solution 47

Example 1

Find dy/dx if $y = x/(2x+5)$.

Set out the factors as for the product rule.

$u = x,\ v = 2x+5, \quad u\prime = 1,\ v\prime = 2.$

$$\frac{dy}{dx} = \frac{1(2x+5) - 2x}{(2x+5)^2} = \frac{5}{(2x+5)^2}.$$

Hints for Example 2

When a fraction equals zero, multiply by its nonzero denominator.

Factorise when one side is zero.

A sine/cosine mixture can be converted to tangent by dividing by cosine, checking it is nonzero.

Example 2

Find the stationary point of $y = \sin x / e^{2x}$, $0 < x < \pi$.

$u = \sin x,\ v = e^{2x}, \quad u\prime = \cos x,\ v\prime = 2e^{2x}.$

$$\frac{dy}{dx} = \frac{e^{2x}\cos x - 2e^{2x}\sin x}{e^{4x}} = 0.$$

$e^{2x}(\cos x - 2\sin x) = 0.$

Since $e^{2x} \neq 0$, $\cos x = 2\sin x$.

$\tan x = \frac{1}{2} \implies x = 0.464\ldots$

$y = \dfrac{\sin x}{e^{2x}} = 0.177\ldots$

Stationary point: $(0.464, 0.177)$, to 3 s.f.

Test Your Understanding — 1/2

Solution 48

Exam-style question 1 (5 marks)

Differentiate $\sin 4x / x^3$, giving your answer in simplest form.

1

$$\frac{dy}{dx} = \frac{4x^3\cos 4x - 3x^2\sin 4x}{x^6}$$

$$= \frac{4x\cos 4x - 3\sin 4x}{x^4}.$$

Test Your Understanding — 2/2

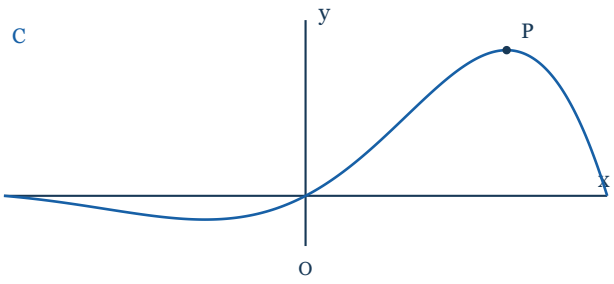
Solution 49

Question 2

Curve C: $y = e^{x\sqrt{3}} \sin 3x$, $-\pi/3 \le x \le \pi/3$.

a. Find the positive x-coordinate of turning point P, as a multiple of π. **(6 marks)**

b. Find the normal at $x = 0$. **(3 marks)**



2a

$\dfrac{dy}{dx} = \sqrt{3}e^{x\sqrt{3}} \sin 3x + 3e^{x\sqrt{3}} \cos 3x.$

Set the derivative to zero:

$e^{x\sqrt{3}}(\sqrt{3}\sin 3x + 3\cos 3x) = 0.$

$\tan 3x = -\sqrt{3}.$

For $0 < x < \pi/3$,

$3x = \dfrac{2\pi}{3} \implies x = \dfrac{2\pi}{9}.$

2b

At $x = 0$, $y = 0$ and $dy/dx = 3$.

Normal gradient $= -1/3$, so

$\dfrac{y-0}{x-0} = -\dfrac{1}{3}, \qquad y = -\dfrac{x}{3}.$

Exercise 9E — Pages 244–245

Solution 50

6. Find the tangent to $y = e^{x/3}/x$ at $(3, e/3)$.

6

Let $u = e^{x/3}$, $v = x$, so $u\prime = \frac{1}{3}e^{x/3}$, $v\prime = 1$.

$\dfrac{dy}{dx} = \dfrac{vu\prime - uv\prime}{v^2}$

$= \dfrac{\frac{x}{3}e^{x/3} - e^{x/3}}{x^2} = \dfrac{e^{x/3}(x/3 - 1)}{x^2}.$

At $x = 3$, $dy/dx = 0$. The tangent is

$y - \dfrac{e}{3} = 0(x - 3), \qquad y = \dfrac{e}{3}.$

Exercise 9E — Pages 244–245 Solution 51

8. Curve C: $x = e^y/(3+2y)$.

a. Find its intercept P on the x-axis. **(1 mark)**

b. Find the normal at P in the form $y = mx + c$, with exact constants. **(6 marks)**

8a

At $y = 0$,

$$x = \frac{e^0}{3} = \frac{1}{3}.$$

Thus $P = (\frac{1}{3}, 0)$.

8b

Let $u = e^y$, $v = 3+2y$; $du/dy = e^y$, $dv/dy = 2$.

$$\frac{dx}{dy} = \frac{e^y(3+2y) - 2e^y}{(3+2y)^2} = \frac{e^y(2y+1)}{(3+2y)^2}.$$

Normal gradient $= -1/(dy/dx) = -dx/dy$.

At P, $m = -e^0(1)/3^2 = -1/9$.

$$y - 0 = -\frac{1}{9}(x - \tfrac{1}{3}) \implies y = -\frac{x}{9} + \frac{1}{27}.$$

$m = -1/9,\ c = 1/27$.

Exercise 9E — Pages 244–245 Solution 52

9. Differentiate $x^4/\cos 3x$ with respect to x.

9

Let $u = x^4$, $v = \cos 3x$.

$u\prime = 4x^3, \quad v\prime = -3\sin 3x.$

$$\frac{dy}{dx} = \frac{vu\prime - uv\prime}{v^2}$$

$$= \frac{4x^3\cos 3x - x^4(-3\sin 3x)}{\cos^2 3x}$$

$$= \frac{x^3(4\cos 3x + 3x\sin 3x)}{\cos^2 3x}.$$

Exercise 9E — Pages 244–245 Solution 53

10. Curve C: $y = e^{2x}/(x-2)^2$, $x \neq 2$.

a. Show that $dy/dx = Ae^{2x}(Bx - C)/(x-2)^3$, finding integers A, B, C.

b. Find the tangent at $x = 1$.

10a

$u = e^{2x},\ v = (x-2)^2, \quad u\prime = 2e^{2x},\ v\prime = 2(x-2).$

$$\frac{dy}{dx} = \frac{2(x-2)^2e^{2x} - 2e^{2x}(x-2)}{(x-2)^4}$$

$$= \frac{2e^{2x}(x-2)[(x-2)-1]}{(x-2)^4}$$

$$= \frac{2e^{2x}(x-3)}{(x-2)^3}.$$

$A = 2,\ B = 1,\ C = 3.$

10b

At $x = 1$, $y = e^2$ and

$$\frac{dy}{dx} = \frac{2e^2(-2)}{-1} = 4e^2.$$

The tangent is

$y - e^2 = 4e^2(x-1), \qquad y = 4e^2x - 3e^2.$

Exercise 9E — Pages 244–245 Solution 54

11. $f(x) = \frac{2x}{x+5} + \frac{6x}{x^2+7x+10}$, $x > 0$.

a. Show that $f(x) = 2x/(x+2)$.

b. Hence find $f\prime(3)$.

11a

$$f(x) = \frac{2x}{x+5} + \frac{6x}{(x+2)(x+5)}$$

$$= \frac{2x(x+2)+6x}{(x+2)(x+5)} = \frac{2x^2+10x}{(x+2)(x+5)}$$

$$= \frac{2x(x+5)}{(x+2)(x+5)} = \frac{2x}{x+2}.$$

Cancellation is valid since $x > 0$, so $x + 5 \neq 0$.

11b

Let $u = 2x$, $v = x+2$; $u\prime = 2$, $v\prime = 1$.

$$f\prime(x) = \frac{2(x+2) - 2x}{(x+2)^2} = \frac{4}{(x+2)^2}.$$

$$f\prime(3) = \frac{4}{5^2} = \frac{4}{25}.$$

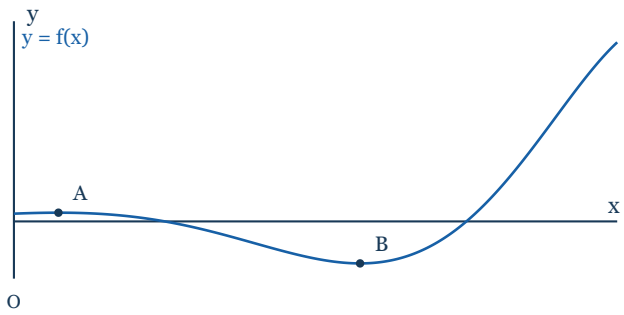
Exercise 9E — Pages 244–245

Solution 55

12. $f(x) = 2\cos 2x / e^{2-x}$, $0 < x < \pi$. The curve has a local maximum at A and a local minimum at B.

a. Show that the x-coordinates of A, B satisfy $\tan 2x = 1/2$. **(4 marks)**

b. Find the range of $f(x)$. **(2 marks)**



12a

Let $u = 2\cos 2x$, $v = e^{2-x}$; $u\prime = -4\sin 2x$, $v\prime = -e^{2-x}$.

$$f\prime(x) = \frac{-4e^{2-x}\sin 2x + 2e^{2-x}\cos 2x}{e^{2(2-x)}}$$

$$= \frac{2e^{2-x}(\cos 2x - 2\sin 2x)}{e^{2(2-x)}}.$$

At stationary points, $\cos 2x - 2\sin 2x = 0$, so $\tan 2x = 1/2$.

12b

$2x = 0.4636\ldots,\ 3.6052\ldots$

$x = 0.2318\ldots,\ 1.8026\ldots$.

The second value is the minimum B.

The range runs from the attained minimum to the excluded right endpoint value:

$f(1.8026\ldots) \le y < f(\pi)$.

Numerically the endpoints are $-1.47\ldots$ and $6.26\ldots$ (3 s.f.).

More exactly, with $b = (\arctan(1/2) + \pi)/2$, $2e^{b-2}\cos(2b) \le y < 2e^{\pi-2}$.

Extension — Stationary Points

Solution 56

STEP I 2010, Q2

The curve $y = \frac{x-a}{x-b}e^x$, with constants a, b, has two stationary points. Show that $a - b < 0$ or $a - b > 4$.

Differentiate

Use the product rule, then the quotient rule:

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{x-a}{x-b}\right)e^x + \frac{x-a}{x-b}e^x$$

$$= \frac{(x-b)-(x-a)}{(x-b)^2}e^x + \frac{x-a}{x-b}e^x$$

$$= \frac{a-b+(x-a)(x-b)}{(x-b)^2}e^x$$

$$= \frac{x^2-(a+b)x+ab+a-b}{(x-b)^2}e^x.$$

Discriminant

At stationary points the numerator is zero:

$x^2 - (a+b)x + ab + a - b = 0$.

Two distinct real roots require a positive discriminant:

$(a+b)^2 - 4(ab+a-b) > 0$

$a^2 - 2ab + b^2 - 4a + 4b > 0$

$(a-b)^2 - 4(a-b) > 0$

$(a-b)(a-b-4) > 0$.

A sketch of $z(z-4)$, or a sign table with $z = a-b$, gives

$a - b < 0 \quad$ or $\quad a - b > 4$.

Differentiating Other Trigonometric Functions

Solution 57

Differentiate $y = \tan x$.

Use the definition of tangent, then the quotient rule.

$$y = \frac{\sin x}{\cos x}, \quad u = \sin x, \quad v = \cos x.$$
$$u\prime = \cos x, \quad v\prime = -\sin x.$$
$$\frac{dy}{dx} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

More generally

$$\frac{d}{dx}\tan kx = k\sec^2 kx.$$

Tip

This is in the formula booklet, but memorising it saves exam time.

Differentiate $y = \sec x$.

$$y = \frac{1}{\cos x} = (\cos x)^{-1}.$$
$$\frac{dy}{dx} = -(\cos x)^{-2}(-\sin x)$$
$$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x}\frac{\sin x}{\cos x} = \sec x \tan x.$$

Tip

The quotient rule would work too; the chain rule is often simpler when the numerator is constant.

More generally,

$$\frac{d}{dx}\sec kx = k\sec kx \tan kx.$$

Further Trigonometric Derivatives

Solution 58

$$\frac{d}{dx}\tan x = \sec^2 x.$$
$$\frac{d}{dx}\sec x = \sec x \tan x.$$
$$\frac{d}{dx}\cot x = -\operatorname{cosec}^2 x.$$
$$\frac{d}{dx}\operatorname{cosec} x = -\operatorname{cosec} x \cot x.$$

Memorisation tip

Adding "co" to the left-hand function adds "co" to the factors in its derivative, and changes the sign.

Example 1

Differentiate $y = \operatorname{cosec} 2x/x^2$.

$$u = \operatorname{cosec} 2x, \quad v = x^2,$$
$$u\prime = -2\operatorname{cosec} 2x \cot 2x, \quad v\prime = 2x.$$
$$\frac{dy}{dx} = \frac{-2x^2 \operatorname{cosec} 2x \cot 2x - 2x \operatorname{cosec} 2x}{x^4}.$$

Example 2

Differentiate $y = \sec^3 x$.

$$y = (\sec x)^3$$
$$\frac{dy}{dx} = 3(\sec x)^2 \sec x \tan x = 3\sec^3 x \tan x.$$

Observation

The power of secant stays the same when differentiating. This will matter in integration.

Test Your Understanding So Far

Solution 59

Edexcel C3 June 2013(R), Q5b (3 marks)

Show that $\frac{d}{dx}(\sec^2 3x)$ can be written as $u(\tan 3x + \tan^3 3x)$, where u is constant.

Solution

$$\frac{d}{dx}(\sec 3x)^2 = 2\sec 3x(3\sec 3x \tan 3x)$$

$$= 6\sec^2 3x \tan 3x$$

$$= 6(1 + \tan^2 3x)\tan 3x$$

$$= 6(\tan 3x + \tan^3 3x).$$

$$u = 6.$$

Using the Reciprocal Derivative

Solution 60

When x is given in terms of y, we may want dy/dx in terms of x. Choose an appropriate trigonometric identity:

$\sin^2 x + \cos^2 x \equiv 1$, $\quad 1 + \tan^2 x \equiv \sec^2 x$.

Example

Given $x = \tan y$, express dy/dx in terms of x.

Working

$$\frac{dx}{dy} = \sec^2 y, \qquad \frac{dy}{dx} = \frac{1}{\sec^2 y}.$$

Since $1 + \tan^2 y = \sec^2 y$,

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}.$$

Choosing the useful identity

Writing $dy/dx = \cos^2 y$ is correct, but it does not yet express the answer in terms of the known $\tan y = x$.

The identity $1 + \tan^2 y = \sec^2 y$ makes that substitution directly.

Further Examples — Inverse Trigonometric Functions

Solution 61

Show that $y = \arcsin x$ gives $dy/dx = 1/\sqrt{1-x^2}$. Use $\sin^2 y + \cos^2 y = 1$ to express $\cos y$ in terms of $x = \sin y$.

$y = \arcsin x \implies x = \sin y.$

$\dfrac{dx}{dy} = \cos y, \qquad \dfrac{dy}{dx} = \dfrac{1}{\cos y}.$

For $-1 < x < 1$, the principal arcsine gives $-\pi/2 < y < \pi/2$, so $\cos y > 0$.

$\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-\sin^2 y}} = \dfrac{1}{\sqrt{1-x^2}}.$

Given $y = \arcsin(x^2)$, find dy/dx.

Example

By the chain rule,

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x^2)^2}}(2x) = \frac{2x}{\sqrt{1-x^4}}.$$

Formula booklet

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}},$$
$$\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}},$$
$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}.$$

Test Your Understanding — Reciprocal and Chain Rules

Solution 62

Edexcel C3 January 2011, Q8b,c

Given $x = \sec 2y$:

b. Find dx/dy in terms of y. **(2 marks)**

c. Hence find dy/dx in terms of x. **(4 marks)**

b

$$\frac{dx}{dy} = 2\sec 2y \tan 2y.$$

$y \neq (2n+1)\pi/4,\ n \in \mathbb{Z}.$

c

$$\frac{dy}{dx} = \frac{1}{2\sec 2y \tan 2y} = \frac{1}{2x\tan 2y}.$$

$\tan^2 2y = \sec^2 2y - 1 = x^2 - 1.$

On a branch with $\tan 2y > 0$,

$$\frac{dy}{dx} = \frac{1}{2x\sqrt{x^2-1}}.$$

On a branch with $\tan 2y < 0$, use the negative square root instead; the derivative is not finite at $x = \pm 1$.

Textbook

Given $y = \arctan\frac{1-x}{1+x}$, find dy/dx. Use the full chain-rule method.

Textbook

Let $u = (1-x)/(1+x)$.

$$\frac{du}{dx} = \frac{-(1+x)-(1-x)}{(1+x)^2} = -\frac{2}{(1+x)^2}.$$

$$y = \arctan u \implies \frac{dy}{du} = \frac{1}{1+u^2}.$$

$$\frac{dy}{dx} = \frac{1}{1+u^2}\left(-\frac{2}{(1+x)^2}\right)$$

$$= -\frac{2}{(1+x)^2+(1-x)^2} = -\frac{1}{1+x^2}, \quad x \neq -1.$$

Exercise 9F — Pages 249–251 Solution 63

5. Curve C: $y = 1/(\cos x \sin x)$, for $0 < x < \pi$ where defined.

a. Find dy/dx.

b. Determine the number of stationary points.

5a

$y = \sec x \operatorname{cosec} x.$

Let $u = \sec x$, $v = \operatorname{cosec} x$, so

$u\prime = \sec x \tan x, \quad v\prime = -\operatorname{cosec} x \cot x.$

$$\frac{dy}{dx} = \sec x(-\operatorname{cosec} x \cot x) + \operatorname{cosec} x(\sec x \tan x)$$

$$= -\frac{\cos x}{\cos x \sin x \sin x} + \frac{\sin x}{\sin x \cos x \cos x}$$

$$= -\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}.$$

5b

At stationary points,

$$\frac{1}{\cos^2 x} - \frac{1}{\sin^2 x} = 0$$

$\tan^2 x = 1 \implies \tan x = \pm 1.$

Within $0 < x < \pi$, the solutions are $x = \pi/4$ and $3\pi/4$.

There are **two stationary points**.

Exercise 9F — Pages 249–251 Solution 64

6. Show that $y = \sec x$ gives $dy/dx = \sec x \tan x$.

6

$$y = \frac{1}{\cos x}.$$

Let $u = 1$, $v = \cos x$, so $u\prime = 0$, $v\prime = -\sin x$.

$$\frac{dy}{dx} = \frac{0 \cos x - 1(-\sin x)}{\cos^2 x}$$

$$= \frac{\sin x}{\cos^2 x} = \sec x \tan x.$$

7. Show that $y = \cot x$ gives $dy/dx = -\operatorname{cosec}^2 x$.

7

$$y = \frac{1}{\tan x}.$$

Let $u = 1$, $v = \tan x$, so $u\prime = 0$, $v\prime = \sec^2 x$.

$$\frac{dy}{dx} = \frac{0 \tan x - \sec^2 x}{\tan^2 x} = -\frac{\sec^2 x}{\tan^2 x}$$

$$= -\frac{1/\cos^2 x}{\sin^2 x/\cos^2 x} = -\frac{1}{\sin^2 x} = -\operatorname{cosec}^2 x.$$

The formula extends to every point where $\cot x$ is defined.

STEP I 2011, Q3

Prove the identity

$4\sin\theta\sin(\pi/3-\theta)\sin(\pi/3+\theta)=\sin 3\theta.$ (*)

i. By differentiating (*) or otherwise, show that

$$\cot\frac{\pi}{9}-\cot\frac{2\pi}{9}+\cot\frac{4\pi}{9}=\sqrt{3}.$$

ii. Set $\theta=\pi/6-\phi$ in (*), or use another method, to obtain a similar identity for $\cos 3\theta$. Deduce that

$\cot\theta\cot(\pi/3-\theta)\cot(\pi/3+\theta)=\cot 3\theta.$

Show that

$$\operatorname{cosec}\frac{\pi}{9}-\operatorname{cosec}\frac{5\pi}{9}+\operatorname{cosec}\frac{7\pi}{9}=2\sqrt{3}.$$

Prove $4\sin\theta\sin(\pi/3-\theta)\sin(\pi/3+\theta)=\sin 3\theta$.

Use two factor formulae:

$2\sin A\sin B=\cos(A-B)-\cos(A+B)$,

$2\sin A\cos B=\sin(A+B)+\sin(A-B)$.

These follow by expanding the right-hand sides using addition formulae and collecting terms.

Proof

First take $A=\pi/3-\theta$, $B=\pi/3+\theta$:

$4\sin\theta\sin(\pi/3-\theta)\sin(\pi/3+\theta)$

$=2\sin\theta[\cos(-2\theta)-\cos(2\pi/3)]$

$=2\sin\theta(\cos 2\theta+\frac{1}{2})$

$=2\sin\theta\cos 2\theta+\sin\theta.$

Now use the second formula with $A=\theta$, $B=2\theta$:

$=\sin 3\theta+\sin(-\theta)+\sin\theta=\sin 3\theta.$

Extension Solution — Part 2 of 2 — 1/2

Solution 67

i. Differentiate the identity

A product of several factors is differentiated one factor at a time, keeping the others, then adding.

$(uvwt\cdots)' = u'vwt\cdots + uv'wt\cdots$

$+uvwt'\cdots + \cdots$.

i

Write $A = \pi/3 - \theta$, $B = \pi/3 + \theta$.

$4[\cos\theta \sin A \sin B - \sin\theta \cos A \sin B$

$+ \sin\theta \sin A \cos B] = 3\cos 3\theta$.

Divide by (*), where its sides are nonzero:

$\cot\theta - \cot(\pi/3 - \theta) + \cot(\pi/3 + \theta) = 3\cot 3\theta$.

Set $\theta = \pi/9$ to obtain

$$\cot\frac{\pi}{9} - \cot\frac{2\pi}{9} + \cot\frac{4\pi}{9} = 3\cot\frac{\pi}{3} = \sqrt{3}.$$

Extension Solution — Part 2 of 2 — 2/2

Solution 68

ii

ii. Put $\theta = \pi/6 - \phi$ in (*):

$4\sin(\pi/6 - \phi)\sin(\pi/6 + \phi)\sin(\pi/2 - \phi)$

$= \sin(\pi/2 - 3\phi)$.

First rewrite $\pi/6 - \phi = \pi/2 - (\pi/3 + \phi)$ and $\pi/6 + \phi = \pi/2 - (\pi/3 - \phi)$.

Using $\sin(\pi/2 - A) = \cos A$,

$4\cos(\pi/3 + \phi)\cos(\pi/3 - \phi)\cos\phi = \cos 3\phi$.

Replace ϕ by θ and divide by (*):

$\cot\theta \cot(\pi/3 - \theta)\cot(\pi/3 + \theta) = \cot 3\theta$.

Final identity

For the final result use $\operatorname{cosec} t = \cot(t/2) - \cot t$.

The requested sum becomes

$(\cot 10^\circ - \cot 50^\circ + \cot 70^\circ)$

$-(\cot 20^\circ - \cot 40^\circ + \cot 80^\circ)$.

The differentiated identity at $\theta = \pi/18$ and $\pi/9$ gives $3\sqrt{3} - \sqrt{3} = 2\sqrt{3}$.

Parametric Differentiation

Solution 69

Parametric equations define x, y (and possibly z) in terms of another parameter, such as t.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad dx/dt \neq 0.$$

Textbook example

Find the gradient when $t = 2$ for $x = t^3 + t$, $y = t^2 + 1$, $t \in \mathbb{R}$.

Gradient

$$\frac{dy}{dt} = 2t, \quad \frac{dx}{dt} = 3t^2 + 1, \quad \frac{dy}{dx} = \frac{2t}{3t^2+1}.$$

At $t = 2$, $dy/dx = 4/13$.

Proof

By the chain rule,

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}.$$

Rearrange to obtain the formula.

Find the normal at $\theta = \pi/6$ on $x = 3\sin\theta$, $y = 5\cos\theta$, $0 \leq \theta < 2\pi$.

Normal

$$\frac{dx}{d\theta} = 3\cos\theta, \quad \frac{dy}{d\theta} = -5\sin\theta, \quad \frac{dy}{dx} = -\frac{5}{3}\tan\theta.$$

At $\theta = \pi/6$, $dy/dx = -5/(3\sqrt{3})$,

$$x = \frac{3}{2}, \quad y = \frac{5\sqrt{3}}{2}.$$

The normal is

$$y - \frac{5\sqrt{3}}{2} = \frac{3\sqrt{3}}{5}(x - \tfrac{3}{2}).$$

Test Your Understanding — Parametric Tangent

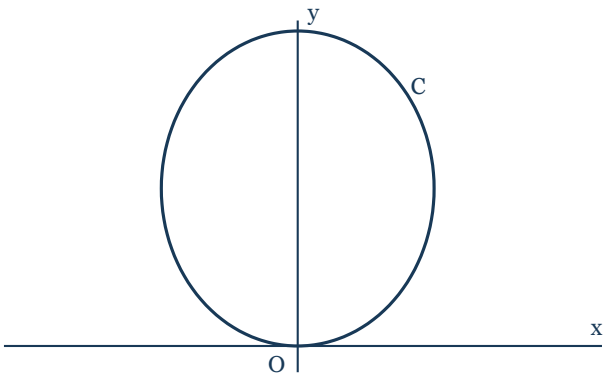
Solution 70

C4 June 2012, Q6

Curve C: $x = \sqrt{3}\sin 2t$, $y = 4\cos^2 t$, $0 \leq t \leq \pi$.

a. Show that $dy/dx = k\sqrt{3}\tan 2t$, finding k. **(5 marks)**

b. Find the tangent at $t = \pi/3$, in the form $y = ax + b$. **(4 marks)**



a

$$\frac{dx}{dt} = 2\sqrt{3}\cos 2t, \qquad \frac{dy}{dt} = -8\cos t\sin t.$$

$$\frac{dy}{dx} = \frac{-8\cos t\sin t}{2\sqrt{3}\cos 2t}$$

$$= \frac{-4\sin 2t}{2\sqrt{3}\cos 2t} = -\frac{2}{3}\sqrt{3}\tan 2t.$$

$k = -2/3$.

b

At $t = \pi/3$, $x = 3/2$, $y = 1$, and

$$m = -\frac{2}{3}\sqrt{3}\tan(2\pi/3) = 2.$$

$$y - 1 = 2(x - \tfrac{3}{2}) \implies y = 2x - 2.$$

Exercise 9G — Pages 252–254 Solution 71

2. a. Find the tangent to $x = 3 - 2\sin t$, $y = t\cos t$, at $t = \pi$.

2a

$$\frac{dx}{dt} = -2\cos t, \quad \frac{dy}{dt} = -t\sin t + \cos t.$$

$$\frac{dy}{dx} = \frac{-t\sin t + \cos t}{-2\cos t} = \frac{t}{2}\tan t - \frac{1}{2}.$$

At $t = \pi$, $x = 3$, $y = -\pi$, $dy/dx = -1/2$.

$y + \pi = -\frac{1}{2}(x - 3)$

$y = -\frac{1}{2}x + \frac{3}{2} - \pi.$

2. b. Find the tangent to $x = 9 - t^2$, $y = t^2 + 6t$, at $t = 2$.

2b

$$\frac{dx}{dt} = -2t, \quad \frac{dy}{dt} = 2t + 6,$$

$$\frac{dy}{dx} = \frac{2t + 6}{-2t} = -\frac{t + 3}{t}.$$

At $t = 2$, $x = 5$, $y = 16$, $dy/dx = -5/2$.

$y - 16 = -\frac{5}{2}(x - 5)$

$2y - 32 = 25 - 5x \implies 2y + 5x = 57.$

Exercise 9G — Pages 252–254 Solution 72

3. a. Find the normal to $x = e^t$, $y = e^t + e^{-t}$, at $t = 0$.

3a

$$\frac{dx}{dt} = e^t, \quad \frac{dy}{dt} = e^t - e^{-t},$$

$$\frac{dy}{dx} = \frac{e^t - e^{-t}}{e^t} = 1 - e^{-2t}.$$

At $t = 0$, the gradient is zero, so the normal is vertical.

Since $x = 1$, $y = 2$, its equation is $x = 1$.

3. b. Find the normal to $x = 1 - \cos 2t$, $y = \sin 2t$, at $t = \pi/6$.

3b

$$\frac{dx}{dt} = 2\sin 2t, \quad \frac{dy}{dt} = 2\cos 2t, \quad \frac{dy}{dx} = \cot 2t.$$

At $t = \pi/6$, $dy/dx = 1/\sqrt{3}$, so normal gradient is $-\sqrt{3}$.

$x = 1 - \cos(\pi/3) = \frac{1}{2}, \quad y = \sin(\pi/3) = \frac{\sqrt{3}}{2}.$

$y - \frac{\sqrt{3}}{2} = -\sqrt{3}(x - \frac{1}{2})$

$y + \sqrt{3}x = \sqrt{3}.$

Exercise 9G — Pages 252–254 Solution 73

4. Find the points of zero gradient on

$$x = \frac{t}{1-t}, \qquad y = \frac{t^2}{1-t}, \quad t \neq 1.$$

You need not classify them as maxima or minima.

4

Using the quotient rule,

$$\frac{dx}{dt} = \frac{(1-t) - t(-1)}{(1-t)^2} = \frac{1}{(1-t)^2},$$

$$\frac{dy}{dt} = \frac{(1-t)2t - t^2(-1)}{(1-t)^2} = \frac{2t - t^2}{(1-t)^2}.$$

$$\frac{dy}{dx} = \frac{2t - t^2}{(1-t)^2} \div \frac{1}{(1-t)^2} = t(2-t).$$

Thus $dy/dx = 0$ at $t = 0$ or 2.

At $t = 0$, $(x, y) = (0, 0)$; at $t = 2$, $(x, y) = (-2, -4)$.

Exercise 9G — Pages 252–254 Solution 74

5. Curve C: $x = e^{2t}$, $y = e^t - 1$, $t \in \mathbb{R}$.

a. Find the tangent at $t = \ln 2$.

b. Show that C has no stationary points.

5a

$$\frac{dx}{dt} = 2e^{2t}, \quad \frac{dy}{dt} = e^t, \quad \frac{dy}{dx} = \frac{e^t}{2e^{2t}} = \frac{1}{2e^t}.$$

At $t = \ln 2$, $x = 4$, $y = 1$, $dy/dx = 1/4$.

$y - 1 = \frac{1}{4}(x - 4) \implies y = \frac{1}{4}x$.

5b

Stationary points require

$$\frac{1}{2e^t} = 0 \implies e^{-t} = 0.$$

This has no real solution, so the curve has no stationary points.

Exercise 9G — Pages 252–254

Solution 75

6. Curve C: $x = (t^2 - 3t - 4)/t$, $y = 2t$, $t > 0$. A tangent l_1 is parallel to $y = x + 5$. Find l_1.

6

$$x = t - 3 - 4t^{-1}, \quad \frac{dx}{dt} = 1 + \frac{4}{t^2}, \quad \frac{dy}{dt} = 2.$$

$$\frac{dy}{dx} = \frac{2}{1 + 4/t^2} = \frac{2t^2}{t^2 + 4}.$$

The parallel line has gradient 1:

$$\frac{2t^2}{t^2 + 4} = 1 \implies t^2 = 4.$$

Since $t > 0$, $t = 2$. Then $x = -3$, $y = 4$.

$$y - 4 = 1(x + 3) \implies l_1 : \ y = x + 7.$$

Exercise 9G — Pages 252–254

Solution 76

7. $x = 2\sin^2 t$, $y = 2\cot t$, $0 < t < \pi/2$.

a. Find dy/dx in terms of t.

b. Find the tangent at $t = \pi/6$.

7a

$$\frac{dx}{dt} = 4\sin t \cos t, \quad \frac{dy}{dt} = -2\operatorname{cosec}^2 t.$$

$$\frac{dy}{dx} = -\frac{2\operatorname{cosec}^2 t}{4\sin t \cos t} = -\tfrac{1}{2}\sec t \operatorname{cosec}^3 t.$$

7b

At $t = \pi/6$, $x = 1/2$, $y = 2\sqrt{3}$,

$$\frac{dy}{dx} = -\tfrac{1}{2} \cdot \frac{2}{\sqrt{3}} \cdot 2^3 = -\frac{8\sqrt{3}}{3}.$$

$$y - 2\sqrt{3} = -\frac{8\sqrt{3}}{3}(x - \tfrac{1}{2})$$

$$\sqrt{3}y - 6 = -8x + 4$$

$$8x + \sqrt{3}y - 10 = 0.$$

Exercise 9G — Pages 252–254 Solution 77

8. Curve C: $x = 4\sin t$, $y = 2\operatorname{cosec} 2t$, $0 < t < \pi$, where defined. Point $A = (2\sqrt{3}, 4\sqrt{3}/3)$.

a. Find t at A.

b. Show that the normal l at A is $9x + 12y - 34\sqrt{3} = 0$.

8a

$4\sin t = 2\sqrt{3} \implies \sin t = \sqrt{3}/2.$

$t = \pi/3$ or $2\pi/3$.

At $t = \pi/3$, $y = 2\operatorname{cosec}(2\pi/3) = 4/\sqrt{3} = 4\sqrt{3}/3$.

The other value gives negative y.

Hence $t = \pi/3$ at A.

8b

$$\frac{dx}{dt} = 4\cos t, \quad \frac{dy}{dt} = -4\operatorname{cosec} 2t \cot 2t.$$

$$\frac{dy}{dx} = -\frac{\cot 2t \operatorname{cosec} 2t}{\cos t}.$$

At $t = \pi/3$:

$$\frac{dy}{dx} = -\frac{(-1/\sqrt{3})(2/\sqrt{3})}{1/2} = \frac{4}{3}.$$

Normal gradient is $-3/4$, so

$$y - \frac{4\sqrt{3}}{3} = -\tfrac{3}{4}(x - 2\sqrt{3})$$

$$12y - 16\sqrt{3} = -9(x - 2\sqrt{3})$$

$$9x + 12y - 34\sqrt{3} = 0.$$

Exercise 9G — Pages 252–254 Solution 78

9. Curve C: $x = t^2 + t$, $y = t^2 - 10t + 5$, $t \in \mathbb{R}$. At P the gradient is 2.

a. Find P.

b. Find the tangent at P.

c. Show that this tangent does not meet C again.

9a

$$\frac{dx}{dt} = 2t + 1, \quad \frac{dy}{dt} = 2t - 10, \quad \frac{dy}{dx} = \frac{2t - 10}{2t + 1}.$$

$$\frac{2t - 10}{2t + 1} = 2 \implies 2t - 10 = 4t + 2 \implies t = -6.$$

$x = (-6)^2 - 6 = 30, \quad y = 36 + 60 + 5 = 101.$

$P = (30, 101)$.

9b $y - 101 = 2(x - 30) \implies y = 2x + 41.$

9c

Substitute the parametric equations:

$t^2 - 10t + 5 = 2(t^2 + t) + 41$

$t^2 + 12t + 36 = 0.$

The discriminant is $12^2 - 4(36) = 0$; the only parameter is $t = -6$. Thus there is no second intersection.

10. Curve C: $x = 2\sin t$, $y = \sqrt{2}\cos 2t$, $0 < t < \pi$.

a. Find dy/dx in terms of t.

b. Find the normal l at A, where $t = \pi/3$, in the form $ax + by + c = 0$.

c. Prove that l meets the curve only at A.

10a

$\dfrac{dx}{dt} = 2\cos t, \quad \dfrac{dy}{dt} = -2\sqrt{2}\sin 2t.$

$\dfrac{dy}{dx} = \dfrac{-2\sqrt{2}\sin 2t}{2\cos t} = -2\sqrt{2}\sin t.$

The quotient calculation assumes $t \neq \pi/2$. At $t = \pi/2$, use $y = \sqrt{2}(1 - x^2/2)$: its tangent gradient at $x = 2$ is $-2\sqrt{2}$.

10b

At $t = \pi/3$, $x = \sqrt{3}$, $y = -\sqrt{2}/2$, $dy/dx = -\sqrt{6}$.

The normal is

$y + \dfrac{\sqrt{2}}{2} = \dfrac{1}{\sqrt{6}}(x - \sqrt{3})$

$x - \sqrt{6}y - 2\sqrt{3} = 0.$

10c

Substitute into the normal:

$2\sin t - \sqrt{6}\sqrt{2}\cos 2t - 2\sqrt{3} = 0$

$\sin t - \sqrt{3}\cos 2t - \sqrt{3} = 0$

$\sin t - \sqrt{3}(1 - 2\sin^2 t) - \sqrt{3} = 0$

$2\sqrt{3}\sin^2 t + \sin t - 2\sqrt{3} = 0$

$(2\sin t - \sqrt{3})(\sqrt{3}\sin t + 2) = 0.$

$\sin t = \sqrt{3}/2$ or $-2/\sqrt{3}$. The second is impossible. The first gives $t = \pi/3$ or $2\pi/3$; both give $(x, y) = (\sqrt{3}, -\sqrt{2}/2) = A$.

$t = 2\pi/3 \implies x = 2\sin(2\pi/3) = \sqrt{3},$

$y = \sqrt{2}\cos(4\pi/3) = -\sqrt{2}/2.$

11. $x = \cos t$, $y = \frac{1}{2}\sin 2t$, $0 \leq t < 2\pi$.

a. Find dy/dx.

b. Find the tangent at A, where $t = \pi/6$.

c. Find the two other tangents l_1, l_2 parallel to the tangent at A.

11a $\dfrac{dx}{dt} = -\sin t, \quad \dfrac{dy}{dt} = \cos 2t, \quad \dfrac{dy}{dx} = -\dfrac{\cos 2t}{\sin t}.$

11b

At $t = \pi/6$, $x = \sqrt{3}/2$, $y = \sqrt{3}/4$,

$dy/dx = -(1/2)/(1/2) = -1.$

$y - \dfrac{\sqrt{3}}{4} = -(x - \dfrac{\sqrt{3}}{2})$

$y = -x + \dfrac{3\sqrt{3}}{4}.$

11c

For gradient -1:

$\dfrac{\cos 2t}{\sin t} = 1 \implies 1 - 2\sin^2 t = \sin t$

$2\sin^2 t + \sin t - 1 = 0$

$(2\sin t - 1)(\sin t + 1) = 0.$

$\sin t = 1/2$ or -1, so $t = \pi/6, 5\pi/6, 3\pi/2$.

The two further points are $(-\sqrt{3}/2, -\sqrt{3}/4)$ and $(0, 0)$.

$l_1 : \ y + \dfrac{\sqrt{3}}{4} = -(x + \dfrac{\sqrt{3}}{2}) \implies y = -x - \dfrac{3\sqrt{3}}{4}.$

$l_2 : \ y = -x.$

Implicit Differentiation

Solution 81

An explicit equation makes y the subject, for example $y = x^2 + 3x$, so we can calculate y directly from x.

An implicit relation may instead be

$x^2 + y^2 = 8x$ or $\cos(x + y) = \sin y$.

Even with $y = x^2$, we differentiate *both* sides with respect to x:

$$\frac{d}{dx}(y) = \frac{d}{dx}(x^2) \quad \Longrightarrow \quad \frac{dy}{dx} = 2x.$$

The derivative of y with respect to x is, by definition, dy/dx.

Two things to remember

1. Differentiate each side of the equation, using the chain rule where needed.
2. Differentiating y with respect to x gives dy/dx.

Examples — Implicit Differentiation

Solution 82

Use the chain rule

"Something squared" differentiates to twice that "something", multiplied by its derivative.

For y^2, multiply $2y$ by dy/dx.

1. $\frac{d}{dx}(y^2)$ — [1] $2y\frac{dy}{dx}$
2. $\frac{d}{dx}(\sin y)$ — [2] $\cos y\frac{dy}{dx}$
3. $\frac{d}{dx}(e^y)$ — [3] $e^y\frac{dy}{dx}$
4. $\frac{d}{dx}(xy)$ — [4] $x\frac{dy}{dx} + y$

In general

$$\frac{d}{dx}f(y) = f'(y)\frac{dy}{dx}.$$

5. $\frac{d}{dx}(e^{x^2y})$ — [5] $(x^2\frac{dy}{dx} + 2xy)e^{x^2y}$
6. $\frac{d}{dx}(\tan(x + y))$ — [6] $(1 + \frac{dy}{dx})\sec^2(x + y)$
7. $\frac{d}{dx}(x^2 + \cos y)$ — [7] $2x - \sin y\frac{dy}{dx}$

Why not just y for the derivative of xy?

Unlike the constant a in ax, y varies with x, so the product rule is needed. This is a very common exam calculation.

Further Examples — Implicit Differentiation

Solution 83

Textbook

Find dy/dx in terms of x, y where $x^3 + x + y^3 + 3y = 6$.

Differentiate both sides: remember that 6 becomes 0. Then make dy/dx the subject.

$$3x^2 + 1 + 3y^2\frac{dy}{dx} + 3\frac{dy}{dx} = 0$$
$$3y^2\frac{dy}{dx} + 3\frac{dy}{dx} = -3x^2 - 1$$
$$\frac{dy}{dx}(3y^2 + 3) = -3x^2 - 1$$
$$\frac{dy}{dx} = \frac{-3x^2 - 1}{3y^2 + 3}.$$

Find dy/dx at $(1, 1)$ where $e^{2x}\ln y = x + y - 2$.

$$2e^{2x}\ln y + e^{2x}\frac{1}{y}\frac{dy}{dx} = 1 + \frac{dy}{dx}.$$

At $x = y = 1$:

$$e^2\frac{dy}{dx} = 1 + \frac{dy}{dx}$$
$$(e^2 - 1)\frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{e^2 - 1}.$$

Tip

Substitute the point early; there is no need to isolate dy/dx first.

Unlike the Year 1 examples, the gradient here can depend on both x and y.

Test Your Understanding — Implicit Curves — 1/2

Solution 84

C4 January 2008, Q5

Curve: $x^3 - 4y^2 = 12xy$.

a. Find the two points where $x = -8$. **(3 marks)**

b. Find the gradient at each. **(6 marks)**

2008 a

$$-512 - 4y^2 = -96y$$
$$y^2 - 24y + 128 = 0$$
$$(y - 16)(y - 8) = 0.$$

Points: $(-8, 16)$, $(-8, 8)$.

2008 b

$$3x^2 - 8y\frac{dy}{dx} = 12x\frac{dy}{dx} + 12y.$$
$$\frac{dy}{dx} = \frac{3x^2 - 12y}{12x + 8y}.$$

The gradients are 0 at $(-8, 16)$, and -3 at $(-8, 8)$.

You can substitute before isolating the derivative.

Test Your Understanding — Implicit Curves — 2/2

Solution 85

C4 June 2014(R), Q3

$x^2 + y^2 + 10x + 2y - 4xy = 10$.

a. Find dy/dx, fully simplified. **(5 marks)**

b. Find y when $dy/dx = 0$. **(5 marks)**

Hint: solve simultaneously with the original equation.

2014 a

$$2x + 2y\frac{dy}{dx} + 10 + 2\frac{dy}{dx} - 4(y + x\frac{dy}{dx}) = 0$$

$$(2y + 2 - 4x)\frac{dy}{dx} = 4y - 2x - 10$$

$$\frac{dy}{dx} = \frac{x + 5 - 2y}{2x - y - 1}.$$

2014 b

$dy/dx = 0 \implies x = 2y - 5$. Substitute:

$$(2y - 5)^2 + y^2 + 10(2y - 5) + 2y - 4(2y - 5)y = 10$$

$$-3y^2 + 22y - 35 = 0$$

$$(3y - 7)(y - 5) = 0 \implies y = \frac{7}{3} \text{ or } 5.$$

Exam note

This type of implicit-differentiation question is common.

Exercise 9H — Pages 255–256

Solution 86

1. Write $u = y^n$ and use the chain rule to show that $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$.

1

$$u = y^n \implies \frac{du}{dy} = ny^{n-1}.$$

$$\frac{d(y^n)}{dx} = \frac{du}{dx} = \frac{du}{dy}\frac{dy}{dx} = ny^{n-1}\frac{dy}{dx}.$$

2. Use the product rule to show that $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$.

2

$$\frac{d(xy)}{dx} = x\frac{dy}{dx} + \frac{dx}{dx}y$$

$$= x\frac{dy}{dx} + 1 \cdot y = x\frac{dy}{dx} + y.$$

Exercise 9H — Pages 255–256

Solution 87

7. Curve C is described by

$2x^2 + 3y^2 - x + 6xy - 1 = 0.$

Find its tangent at $(1, -2)$ in the form $ax + by + c = 0$, with integer constants.

7

$$4x + 6y\frac{dy}{dx} - 1 + 6(x\frac{dy}{dx} + y) = 0$$

$$(6y + 6x)\frac{dy}{dx} = 1 - 6y - 4x$$

$$\frac{dy}{dx} = \frac{1 - 6y - 4x}{6(x + y)}.$$

At $(1, -2)$:

$$\frac{dy}{dx} = \frac{1 - 6(-2) - 4}{6(1 - 2)} = -\frac{3}{2}.$$

$$y + 2 = -\tfrac{3}{2}(x - 1)$$

$$2y + 4 = -3x + 3 \implies 3x + 2y + 1 = 0.$$

Exercise 9H — Pages 255–256

Solution 88

8. Curve C has equation $3^x = y - 2xy$. Find the exact gradient at $(2, -3)$.

8

$$3^x \ln 3 = \frac{dy}{dx} - 2(x\frac{dy}{dx} + y)$$

$$3^x \ln 3 + 2y = (1 - 2x)\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{3^x \ln 3 + 2y}{1 - 2x}.$$

At $x = 2, y = -3$:

$$\frac{dy}{dx} = \frac{9 \ln 3 - 6}{1 - 4} = 2 - 3 \ln 3.$$

Exercise 9H — Pages 255–256

Solution 89

9. Find the exact gradient of
$\ln(y^2) = \frac{1}{2}x\ln(x-1), \quad x > 1,\ y > 0,$
at the point where $x = 4$.

9

$$2\ln y = \tfrac{1}{2}x\ln(x-1)$$
$$\frac{2}{y}\frac{dy}{dx} = \frac{1}{2}\left(\frac{x}{x-1} + \ln(x-1)\right)$$
$$\frac{dy}{dx} = \frac{y}{4}\left(\frac{x}{x-1} + \ln(x-1)\right).$$

At $x = 4$,

$$\ln(y^2) = 2\ln 3 = \ln 9 \implies y^2 = 9.$$

Since $y > 0$, $y = 3$. Hence

$$\frac{dy}{dx} = \frac{3}{4}(\tfrac{4}{3} + \ln 3) = 1 + \frac{3}{4}\ln 3.$$

Exercise 9H — Pages 255–256

Solution 90

10. Curve C satisfies $\sin x + \cos y = 0.5$, with $-\pi < x < \pi$, $-\pi < y < \pi$.

a. Find dy/dx.

b. Find its stationary points.

10a

$$\cos x - \sin y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{\cos x}{\sin y}.$$

10b

For $dy/dx = 0$, $\cos x = 0$, so $x = \pm\pi/2$.

If $x = \pi/2$, $1 + \cos y = 0.5$, hence $\cos y = -0.5$, $y = \pm 2\pi/3$.

If $x = -\pi/2$, $-1 + \cos y = 0.5$, so $\cos y = 1.5$: impossible.

The stationary points are

$(\tfrac{\pi}{2}, \tfrac{2\pi}{3})$, $\quad (\tfrac{\pi}{2}, -\tfrac{2\pi}{3})$.

Exercise 9H — Pages 255–256 Solution 91

11. Curve C: $ye^{-3x} - 3x = y^2$.

a. Find dy/dx in terms of x, y.

b. Show that the tangent at the origin is $y = 3x$.

11a

$$y(-3e^{-3x}) + e^{-3x}\frac{dy}{dx} - 3 = 2y\frac{dy}{dx}$$

$$(e^{-3x} - 2y)\frac{dy}{dx} = 3(ye^{-3x} + 1)$$

$$\frac{dy}{dx} = \frac{3(ye^{-3x} + 1)}{e^{-3x} - 2y}.$$

11b

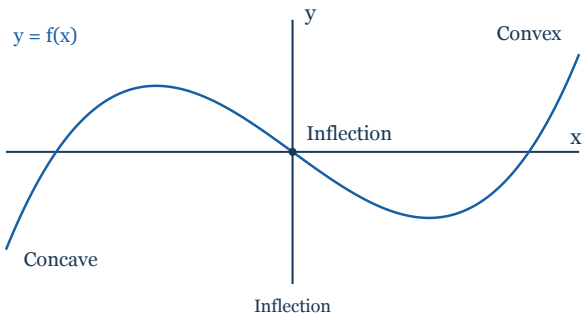
At $(0, 0)$:

$$\frac{dy}{dx} = \frac{3(0e^0 + 1)}{e^0 - 0} = 3.$$

$$y - 0 = 3(x - 0) \implies y = 3x.$$

Using the Second Derivative Solution 92

A point of inflection is where concavity changes: concave to convex, or vice versa. Informally, the curve changes the direction in which it is "swerving".



Concave

The gradient decreases: $f''(x) \leq 0$ on the interval.

Convex

The gradient increases: $f''(x) \geq 0$ on the interval. The gradient may still be negative while increasing.

Inflection

When f'' is continuous, look for $f''(x) = 0$, then check that its sign changes. The gradient's rate of change is zero there, but the gradient itself need not be zero: an inflection need not be stationary.

Memorisation tip

Picture a positive cubic, rising overall. Read "concave" then "convex" from left to right: alphabetical order.

Examples — Concavity and Inflection

Solution 93

1. Find where $f(x) = x^3 + 4x + 3$ is concave.

1

$f'(x) = 3x^2 + 4, \quad f''(x) = 6x.$

$6x \le 0 \implies x \le 0.$

2. Show that $f(x) = e^{2x} + x^2$ is convex for all real x.

2

$f'(x) = 2e^{2x} + 2x, \quad f''(x) = 4e^{2x} + 2.$

Since $e^{2x} > 0$, $f''(x) > 0$ for all real x, so f is convex everywhere.

3. Find the point of inflection of $y = x^3 - 2x^2 - 4x + 5$.

3

$\dfrac{dy}{dx} = 3x^2 - 4x - 4, \quad \dfrac{d^2y}{dx^2} = 6x - 4.$

$6x - 4 = 0 \implies x = 2/3$; the second derivative changes sign there.

$y = (\tfrac{2}{3})^3 - 2(\tfrac{2}{3})^2 - 4(\tfrac{2}{3}) + 5 = \dfrac{47}{27}.$

Point of inflection: $(2/3, 47/27)$.

Exercise 9I — Pages 259–261

Solution 94

9. Given $y = x(3x - 1)^5$:

a. Find dy/dx and d^2y/dx^2.

b. Find the points of inflection.

9a

$\dfrac{dy}{dx} = 15x(3x - 1)^4 + (3x - 1)^5.$

$\dfrac{d^2y}{dx^2} = 15(3x - 1)^4 + 180x(3x - 1)^3 + 15(3x - 1)^4$

$= 30(3x - 1)^4 + 180x(3x - 1)^3$

$= 30(3x - 1)^3(9x - 1).$

9b

$30(3x - 1)^3(9x - 1) = 0 \implies x = \tfrac{1}{3}$ or $\tfrac{1}{9}$.

The second derivative changes sign at both roots: its signs are $+, -, +$ across $1/9, 1/3$.

At $x = 1/9$,

$y = \tfrac{1}{9}(\tfrac{1}{3} - 1)^5 = -\dfrac{32}{2187}.$

At $x = 1/3$, $y = 0$.

Points: $(1/9, -32/2187)$, $(1/3, 0)$.

Exercise 9I — Pages 259–261

Solution 95

11. A curve has equation $y = \frac{1}{3}x^2 \ln x - 2x + 5$, $x > 0$. Show that it is convex for all $x \geq e^{-3/2}$.

11

$$\frac{dy}{dx} = \frac{1}{3}x^2\frac{1}{x} + \frac{2}{3}x\ln x - 2$$

$$= \frac{x}{3} + \frac{2}{3}x\ln x - 2.$$

$$\frac{d^2y}{dx^2} = \frac{1}{3} + \frac{2}{3}(1 + \ln x) = 1 + \frac{2}{3}\ln x.$$

For convexity,

$$1 + \frac{2}{3}\ln x \geq 0 \implies \ln x \geq -\frac{3}{2} \implies x \geq e^{-3/2}.$$

Relating Rates of Change

Solution 96

Find the rate of change of the area A of a circle when $r = 3$ cm, given $dr/dt = 5$ cm s^{-1}.

First represent the statements mathematically.

"Rate of change of area"

Area rate

dA/dt

"Radius changes at a rate of 5"

Radius rate

$dr/dt = 5$

"Area of a circle"

Area

$A = \pi r^2$

Tip

When you see "rate", think differentiation with respect to time.

Build the chain rule

Copy the desired numerator and denominator into the diagonal positions, then fill the remaining positions with the intermediate variable r.

$$\frac{dA}{dt} = \frac{dA}{\square} \times \frac{\square}{dt}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

3 — Calculate

$$\frac{dA}{dt} = 2\pi r \cdot 5 = 2\pi(3)\cdot 5 = 30\pi \text{ cm}^2\text{s}^{-1}.$$

Further Example — Expanding Cylinder

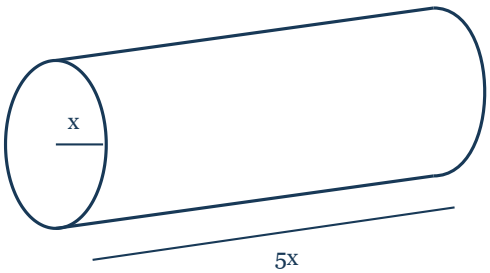
Solution 97

Edexcel C4 June 2008, Q3

A right circular cylindrical metal rod expands as it is heated. At time t seconds its radius is x cm and length $5x$ cm. Its cross-sectional area increases at $0.032\ \text{cm}^2\text{s}^{-1}$.

a. Find dx/dt when $x = 2$, to 3 s.f. **(4 marks)**

b. Find the volume's rate of increase when $x = 2$. **(4 marks)**



It is common for a second part to connect a further pair of rates.

a

$$A = \pi x^2, \quad \frac{dA}{dx} = 2\pi x, \quad \frac{dA}{dt} = 0.032.$$
$$\frac{dx}{dt} = \frac{dA}{dt}\frac{dx}{dA} = 0.032\frac{1}{2\pi x} = \frac{0.016}{\pi x}.$$
At $x = 2$: $dx/dt = 0.002546\ldots \approx 0.00255\ \text{cm s}^{-1}$.

b

$$V = \pi x^2(5x) = 5\pi x^3, \quad \frac{dV}{dx} = 15\pi x^2.$$
$$\frac{dV}{dt} = \frac{dV}{dx}\frac{dx}{dt} = 15\pi x^2\frac{0.016}{\pi x} = 0.24x.$$
At $x = 2$: $dV/dt = 0.48\ \text{cm}^3\text{s}^{-1}$.

Test Your Understanding — Expanding Cube

Solution 98

June 2012, Q2

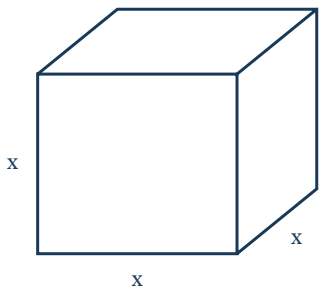
A metal cube expands uniformly as it is heated. At time t seconds, each edge is x cm and the volume is $V\ \text{cm}^3$.

a. Show that $dV/dx = 3x^2$. **(1 mark)**

The volume increases at $0.048\ \text{cm}^3\text{s}^{-1}$.

b. Find dx/dt when $x = 8$. **(2 marks)**

c. Find the surface area's rate of increase when $x = 8$. **(3 marks)**



a

$$V = x^3 \implies \frac{dV}{dx} = 3x^2.$$

b

$$\frac{dx}{dt} = \frac{dx}{dV}\frac{dV}{dt} = \frac{1}{3x^2}(0.048).$$
At $x = 8$:
$$\frac{dx}{dt} = \frac{0.048}{3 \cdot 8^2} = 0.00025\ \text{cm s}^{-1}.$$

c

$$A = 6x^2, \quad \frac{dA}{dx} = 12x.$$
$$\frac{dA}{dt} = \frac{dA}{dx}\frac{dx}{dt} = (12 \cdot 8)(0.00025)$$
$$= 0.024\ \text{cm}^2\text{s}^{-1}.$$

Tip

$dx/dV = 1/(dV/dx)$.

Exercise 9J — Pages 263–264 Solution 99

Note

Setting up differential equations is covered in Integration. Leave Questions 5, 6, 8, 9, 12 and 14 until that topic has been studied.

1. Given $A = \frac{1}{4}\pi r^2$, $dr/dt = 6$, find dA/dt when $r = 2$.

1
$$\frac{dA}{dr} = \frac{1}{2}\pi r, \qquad \frac{dA}{dt} = \frac{dA}{dr}\frac{dr}{dt} = \frac{1}{2}\pi r(6) = 3\pi r.$$
At $r = 2$, $dA/dt = 6\pi$.

2. Given $y = xe^x$, $dx/dt = 5$, find dy/dt when $x = 2$.

2
$$\frac{dy}{dx} = xe^x + e^x = (x+1)e^x.$$
$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} = (x+1)e^x(5).$$
At $x = 2$, $dy/dt = 5(3)e^2 = 15e^2$.

Exercise 9J — Pages 263–264 Solution 100

3. Given $r = 1 + 3\cos\theta$, $d\theta/dt = 3$, find dr/dt when $\theta = \pi/6$.

3
$$\frac{dr}{d\theta} = -3\sin\theta, \qquad \frac{dr}{dt} = \frac{dr}{d\theta}\frac{d\theta}{dt} = -9\sin\theta.$$
At $\theta = \pi/6$:
$$\frac{dr}{dt} = -9\sin(\pi/6) = -\frac{9}{2}.$$

4. Given $V = \frac{1}{3}\pi r^3$, $dV/dt = 8$, find dr/dt when $r = 3$.

4
$$\frac{dV}{dr} = \pi r^2 \implies \frac{dr}{dV} = \frac{1}{\pi r^2}.$$
$$\frac{dr}{dt} = \frac{dr}{dV}\frac{dV}{dt} = \frac{1}{\pi r^2}(8) = \frac{8}{\pi r^2}.$$
At $r = 3$, $dr/dt = 8/(9\pi)$.

Extension — Cone Optimisation

Solution 101

STEP I 2009, Q5

A right circular cone has radius r, height h, slant height l, volume V and curved surface area A:

$$V = \frac{1}{3}\pi r^2 h, \qquad A = \pi r l.$$

i. With A fixed, r is chosen so that V is stationary. Show that $A^2 = 3\pi^2 r^4$ and $l = \sqrt{3}r$.

ii. Instead, V is fixed and r is chosen so that A is stationary. Find h in terms of r.

ii

Solution to ii

$l^2 = h^2 + r^2 = A^2/(\pi^2 r^2)$, and $h = 3V/(\pi r^2)$.

$A^2 = \pi^2 r^2 (h^2 + r^2)$

$$= \pi^2 r^2 \left(\frac{9V^2}{\pi^2 r^4} + r^2 \right) = \frac{9V^2}{r^2} + \pi^2 r^4.$$

Differentiate with V fixed:

$$2A\frac{dA}{dr} = -\frac{18V^2}{r^3} + 4\pi^2 r^3.$$

At a stationary value, $4\pi^2 r^6 = 18V^2$, so $2\pi r^3 = 3\sqrt{2}V$.

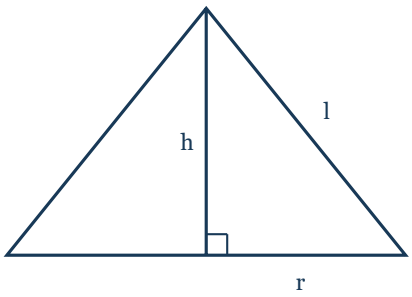
$$h = \frac{3V}{\pi r^2} = \frac{2\pi r^3/\sqrt{2}}{\pi r^2} = \sqrt{2}r.$$

Extension Solution — Fixed Surface Area

Solution 102

Solution to i

With A fixed, $l = A/(\pi r)$. A side view of the cone gives $l^2 = h^2 + r^2$.



Geometry

$$l^2 = h^2 + r^2 = \frac{A^2}{\pi^2 r^2}$$

$$h^2 = \frac{A^2}{\pi^2 r^2} - r^2.$$

Stationary volume

$$V = \tfrac{1}{3}\pi r^2 h \implies V^2 = \tfrac{1}{9}\pi^2 r^4 h^2$$

$$= \frac{\pi^2 r^4}{9}\left(\frac{A^2}{\pi^2 r^2} - r^2 \right)$$

$$= \frac{1}{9}(A^2 r^2 - \pi^2 r^6).$$

Working with V^2 avoids square roots.

$$2V\frac{dV}{dr} = \frac{1}{9}(2A^2 r - 6\pi^2 r^5).$$

At a stationary value, $dV/dr = 0$, so

$2A^2 r - 6\pi^2 r^5 = 0$.

As $r \neq 0$, $A^2 = 3\pi^2 r^4$. Finally,

$$l^2 = \frac{A^2}{\pi^2 r^2} = \frac{3\pi^2 r^4}{\pi^2 r^2} = 3r^2 \implies l = \sqrt{3}r.$$